

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Its structure contains 32 faces☒ Has a relative molecular mass of 60☐ Is an allotrope of carbon☒ Has a giant ionic structure☐ is a hydrocarbon compound☐ It is a smart material☐ Is 1 x 10²⁵ time smaller than a football☒ award (2) for all 3 correct award (1) for any 2 correct award (1) for 3 correct and 1 incorrect if four boxes ticked award (0) for 2 correct and 2 incorrect if four boxes ticked	2			2		
	(b)	(i)		90 (2) award (1) for 180 or for the total number of sides on all pentagons or all hexagons i.e. (12 × 5) / 60 / (20 × 6) / 120 no ECF if incorrect number of sides calculated in first step	1	1		2	2	
		(ii)		6.97 × 10 ⁻²⁸ (3) award (2) for correct calculation of 0.697 / 0.69655 if incorrect answer award (1) for correct identification of radius as 0.55 if final calculation is totally incorrect award (2) for correct answers using the diameter instead of radius (5.57 × 10 ⁻²⁷) award (1) for answer not given in standard form (5.57) accept correct answers using π value from the calculator	1	2		3	3	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			large surface area	1			1		
	(d)			do not know the long term effects (of nano-particles)	1			1		
	(e)			good conductor it is similar to graphite / has delocalised electrons / only bonded to three other carbon atoms (1) not good conductor it is (simple) molecular / exists as discrete molecules / delocalised electrons trapped within the ball structure (1) award (1) if both reasons given but not linked to good / not good conductor			2	2		
				Question 5 total	6	3	2	11	5	0